

A voltage source $\underline{E} = 12j[\dots 1.]$ powers a circuit, determining a current flow through this one of value $\underline{I} = 3 + 3j[\dots 2.]$. Compute the impedance of this circuit, $\underline{Z}$ 3, and explain the significance of the real part 4 and the imaginary part of the impedance 5. The impedance is obtained by connecting in series two elements: identify them 6, compute their values 7 and state their measurements units 8. Find the sinusoidal expression of the current $i(t)$ 9, and compute its value when $t=5[\text{ms}]$ 10 if the circuit works on the frequency $f=40[\dots 11..]$. State the measurement units for the following quantity: $\bar{D}$ 12.

Laural.Darabant@gmail.com

1. Volt, V - 0,8

2. Ampere, A - 0,8

$$3. \underline{Z} = \frac{\underline{U}}{\underline{I}} = \frac{12j}{3+3j} = \frac{12e^{j\frac{\pi}{2}}}{3\sqrt{2}e^{j\frac{\pi}{4}}} = 2\sqrt{2}e^{j(\frac{\pi}{2}-\frac{\pi}{4})} = 2\sqrt{2}e^{j\frac{\pi}{4}} = 2 + 2j - 3$$

4. R - resistance - 2

5. X - reactance - 2

6. Resistance and coil - 2

$$7. R=2 \text{ and } \omega L = X = 2 \rightarrow L = \frac{2}{\omega} = \frac{2}{2\pi \cdot 40} = 7,95[mH] - 2$$

8. Ω - Ohm and H - Henry - 1,6

$$9. i(t) = 6 \sin\left(\omega t + \frac{\pi}{4}\right) - 1,2$$

$$10. i(5ms) = 6 \sin\left(2\pi \cdot 40 \cdot 5 \cdot 10^{-3} + \frac{\pi}{4}\right) \cong 6 \sin(\pi \cdot 0,65) \cong 5,34[A] - 3$$

11. Hz - Hertz - 0,8

12. C/m² - Coulomb per square meter 0,8